

Notes on Interdependent Value Auctions (Slides 13–18)

1. Symmetric Model

We assume all signals X_i lie in the same interval $[0, \omega]$ and that the joint density $f(x_1, \dots, x_n)$ is symmetric. The valuation function takes the form

$$V_i = u(X_i, X_{-i}),$$

where u is common across bidders, strictly increasing in its first coordinate, and symmetric in the remaining $n - 1$ coordinates. Signals are assumed to be *affiliated*, in the Milgrom–Weber sense.

2. The Key Object $\nu(x, y)$

Define

$$\nu(x, y) = E[V_1 \mid X_1 = x, Y_1 = y], \quad Y_1 = \max_{j \neq 1} X_j.$$

This is bidder 1's conditional expected value given own signal x and the highest competing signal y . The function ν is non-decreasing in both arguments and strictly increasing in x . In a symmetric equilibrium, the relevant case is the *diagonal* $\nu(x, x)$.

3. Second-Price Auction

Proposition. In a symmetric equilibrium of the SPA,

$$\beta_{II}(x) = \nu(x, x).$$

Sketch of Proof. Let bidders $j \neq 1$ follow β_{II} . Bidder 1 with signal x who bids b obtains expected utility

$$U(b, x) = \int_0^{\beta^{-1}(b)} (\nu(x, y) - \nu(y, y)) g(y \mid x) dy,$$

where $g(\cdot \mid x)$ is the pdf of $Y_1 \mid X_1 = x$. Since

$$\nu(x, y) - \nu(y, y) > 0 \quad \text{for } y < x, \quad \nu(x, y) - \nu(y, y) < 0 \quad \text{for } y > x,$$

the maximizer solves $\beta^{-1}(b) = x$, hence $b = \nu(x, x)$.

4. First-Price Auction: Expected Utility

Let others follow a strictly increasing differentiable strategy β . If bidder 1 has signal x and bids as if she were type z , she wins when $Y_1 < z$, so her payoff is

$$U(z, x) = \int_0^z (\nu(x, y) - \beta(z)) g(y \mid x) dy.$$

5. Differentiating Using Leibniz' Rule

Let

$$F(y, z) = (\nu(x, y) - \beta(z))g(y | x).$$

Leibniz' rule gives

$$\frac{\partial}{\partial z} \int_0^z F(y, z) dy = F(z, z) + \int_0^z \frac{\partial F}{\partial z}(y, z) dy.$$

Compute terms:

$$F(z, z) = (\nu(x, z) - \beta(z))g(z | x),$$

and

$$\frac{\partial F}{\partial z}(y, z) = -\beta'(z)g(y | x),$$

so

$$\int_0^z -\beta'(z)g(y | x) dy = -\beta'(z)G(z | x).$$

Hence,

$$\frac{\partial U}{\partial z} = (\nu(x, z) - \beta(z))g(z | x) - \beta'(z)G(z | x).$$

At symmetric equilibrium, set $z = x$, giving the ODE:

$$\beta'(x) = (\nu(x, x) - \beta(x)) \frac{g(x | x)}{G(x | x)}.$$

6. Boundary Condition

Since $\nu(0, 0) = 0$, bidding anything other than 0 at type 0 is strictly dominated. Thus:

$$\beta(0) = 0.$$

7. Integrating Factor Trick

We solve the linear ODE:

$$\beta'(y) + p(y)\beta(y) = q(y), \quad p(y) = \frac{g(y | y)}{G(y | y)}, \quad q(y) = \nu(y, y) \frac{g(y | y)}{G(y | y)}.$$

We seek an integrating factor $L(y | x)$ such that

$$\frac{\partial L(y | x)}{\partial y} = p(y)L(y | x),$$

and normalized so that

$$L(x | x) = 1.$$

Solving,

$$L(y | x) = \exp\left(-\int_y^x p(s) ds\right) = \exp\left(-\int_y^x \frac{g(s | s)}{G(s | s)} ds\right).$$

This special two-argument form treats y as the variable of integration and x as the fixed evaluation point. The normalization $L(x | x) = 1$ ensures that the product $\beta(x)L(x | x)$ evaluates to $\beta(x)$ directly.

8. Solving the Differential Equation

Multiplying the ODE by $L(y | x)$:

$$L(y | x)\beta'(y) + p(y)L(y | x)\beta(y) = q(y)L(y | x).$$

The left-hand side becomes the total derivative:

$$\frac{\partial}{\partial y}(\beta(y)L(y | x)) = q(y)L(y | x).$$

Integrate from 0 to x :

$$\beta(x)L(x | x) - \beta(0)L(0 | x) = \int_0^x q(y)L(y | x) dy.$$

Using $L(x | x) = 1$ and $\beta(0) = 0$,

$$\beta(x) = \int_0^x q(y)L(y | x) dy.$$

Thus the equilibrium strategy in the FPA is:

$$\boxed{\beta_I(x) = \int_0^x \nu(y, y) dL(y | x)}.$$

9. Interpretation of the Integrating Factor

The integrating factor is written as $L(y | x)$ rather than the standard $M(y)$ because:

- The ODE is solved in the variable y , but the final evaluation point is x .
- We require a normalization $L(x | x) = 1$ to obtain a clean expression for $\beta(x)$.
- Treating x as a fixed parameter and y as the variable naturally leads to the expression

$$L(y | x) = \exp\left(-\int_y^x p(s) ds\right).$$

- This ensures the boundary condition at 0 and the correct evaluation at x .